MILITARY SPECIFICATION

# SEMICONDUCTOR DEVICES, THYRISTOR, REVERSE BLOCKING, SILICON, TYPES 2N1792 2N1793, 2N1795, 2N1797, 2N1798, 2N1799, 2N1800, 2N1805, 2N1806, 2N1910, 2N1911, 2N1913, 2N1915, 2N1916, 2N2031, JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon thyristors. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figures 1 (TO-94) and 2 (TO-83).

1.3 Maximum ratings.

| $I_{T(AV)}$ 1/ | $I_{TSM}$ 2/ | $-V_{GM}$ | $V_{GM}$ | $I_{GM}$ | $P_{GM}$ | $P_{G(AV)}$ | Stud torque | Operating altitude | $T_{op}$ (case temperture) | $T_{STG}$ and $T_J$ | $R_{\theta JC}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Adc | A(pk)(1 cycle) | v(pk) | v(pk) | A(pk) | W | W | lb-in | mm.$Hg$ | °C | °C | °C/W |
| 50 | 1000 | 5 | 10 | 2 | 5 | 0.5 | 150 | 15 | -40 to +125 | -40 to +150 | 0.4 |

1/ This average on-state current is for a maximum allowable case temperture of 83°C and 180 electrical degrees of half sine wave conduction. For other operating conditions see figure 3.

2/ Surge rating is nonrecurrent and applies only with device in the conducting state.

1.3.1 Individual ratings.

| Types | $V_{D(RMS)}$ | $V_{RRM}$ 1/ | $V_{DRM}$ 1/ | $V_{RSM}$ 2/ |
|---|---|---|---|---|
| | $V_{RMS}$ | V(pk) | V(pk) | V(pk) |
| 2N2031 | 35 | 50 | 50 | 75 |
| 2N1792, 2N1910 | 35 | 50 | 50 | 75 |
| 2N1793, 2N1911 | 70 | 100 | 100 | 150 |
| 2N1795, 2N1913 | 140 | 200 | 200 | 300 |
| 2N1797, 2N1915 | 210 | 300 | 300 | 400 |
| 2N1798, 2N1916 | 280 | 400 | 400 | 500 |
| 2N1799, 2N1805 | 350 | 500 | 500 | 650 |
| 2N1800, 2N1806 | 420 | 600 | 600 | 720 |

1/ Values apply for zero or negative gate voltage ($V_G$).

2/ Transient inverse voltage, nonrecurrent, t = 5.0 ms max.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Washington, DC 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



FSC 5961

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1.4 Primary electrical characteristics at $T_A$ = 25°C (unless otherwise specified).

| Limits | $V_{TM}$ 1/ | $I_H$ 1/ | dv/dt 1/ | $V_{GT}$ 1/ | $I_{GT}$ 1/ | $t_q$ 1/ $T_C$ = 100°C |
|---|---|---|---|---|---|---|
| | v(pk) | mAdc | V/µs | Vdc | mAdc | µs |
| Min | --- | --- | 200 | 0.25 2/ | 5 | --- |
| Max | 2.1 | 40 | --- | 3.0 | 70 3/ | 40 |

1/ See conditions column for the applicable group A test.

2/ This value also applies at $T_A$ = +125°C.

3/ This value also applies at $T_A$ = -40°C.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figures 1 and 2 herein.

3.4 Marking. Marking shall be in accordance with MIL-S-19500, except that the country of origin may be omitted at the option of the manufacturer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.3 Screening (JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II) and as specified herein. The following electrical measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.



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| Screen (see table II of MIL-S-19500) | Measurements JANTX and JANTXV levels |
|---|---|
| 3 | $T_{Low}$ = -40°C; $T_{high}$ = +150°C |
| 4 | 2500 G (leads removed or secured) |
| 9 | Not required |
| 10 | Not required |
| 11 | $I_{RRM1}$, $I_{DRM1}$, $V_{GT1}$, and $V_{TM}$ |
| 12 | Test condition A (ac blocking voltage); $T_C$ = 125°C; $V_{RM}$ = $V_{RRM}$ (see 1.3.1), $V_{DM}$ = $V_{DRM}$ (see 1.3.1); $R_{GK}$ = ∞ |
| 13 | Subgroup 2 of table 1 herein, $\Delta I_{DRM1}$ = 100% of initial value or 1 mA(pk), whichever is greater. $\Delta I_{RRM1}$ = 100% of initial value or 1 mA(pk), whichever is greater. |

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein. All testing shall be conducted on devices prior to attachment of external leads or terminals. After attachment of leads or terminals, the tests of subgroups 1 and 2 of table I shall be performed prior to shipment.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. End point electrical measurements shall be in accordance with the applicable steps of table IV herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb of MIL-S-19500, and table II herein. Electrical measurements shall be in accordance with the applicable steps of table IV herein. Subgroups 3 and 6 of table II shall be performed on a sample from the sublot containing the highest voltage rated devices in the lot.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements shall be in accordance with the applicable steps of table IV herein. Subgroups 5 and 6 of table III shall be performed on a sample from the sublot containing the highest voltage rated devices in the lot to accept that type and all lower voltage rated devices. In the event subsequent lots contain a higher voltage type, that type shall be subjected to the tests of subgroups 5 and 6 prior to the acceptance of the lot.



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4.5 Methods of inspection. Methods of inspection shall be as specified in appropriate tables and as follows:

4.5.1 Intermittent life test. The thyristors shall be operated in a single phase circuit as shown on figure 4 with 175 ±5 degree conduction angle. Cell stud temperature shall be maintained at 80° ±5°C as measured with a thermocouple rigidly attached to a cell hex. Peak input voltage from a 60 Hz sinusoidal blocking voltage power supply shall be equal to rated blocking voltage. Average forward current shall be 50 amperes per cell from the 60 Hz sinusoidal forward current supply. The blocking voltage supply shall be phase delayed 90 electrical degrees with respect to the forward current supply in order to impose both forward and reverse blocking duty on the cells under test. The forward current supply consists of a low voltage transformer with a means of adjusting the secondary voltage. The blocking voltage supply consists of a high voltage transformer with a synchronous switch connected in series with one of the secondary transformer leads. The synchronous switch is set to conduct for 170 ±5 electrical degrees conduction during the interval when terminal A is negative with respect to terminal C. A minimum of 2 and a maximum of 5 electrical degrees off time shall be allowed after forward current conduction before blocking voltage is applied by the synchronous switch. The gate firing pulses shall be delivered from a firing circuit with an open circuit voltage of 7.0 ±3.0 volts and a short circuit current of 1.2 ±0.8 amperes. The gate pulse average power input shall not exceed 0.5 watt. The gate firing supply shall be in phase with the cell forward current supply.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Air Force - 11, 19, 85, 99
DLA - ES
NASA - NA

User activities:
Army - SM
Navy - AS, CG, MC, OS, SH

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5961-0810)



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Physical (aged)

TABLE I. Group A inspection.

| Inspection | MIL-STD-750 Method | MIL-STD-750 Conditions | LTPD JANTX JANTXV | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|
| Subgroup 1 | | | 5 | | | | |
| Visual and mechanical inspections | 2071 | | | | | | |
| Subgroup 2 | | | 5 | | | | |
| Reverse blocking current | 4211 | AC method, bias condition D, f = 60 Hz, $V_{RM}$ = $V_{RRM}$ (see 1.3.1) | | $I_{RRMI}$ | --- | 5 | mA(pk) |
| Forward blocking current | 4206 | AC method, bias condition D, f = 60 Hz, $V_{DM}$ = $V_{DRM}$ (see 1.3.1) | | $I_{DRM1}$ | --- | 5 | mA(pk) |
| Gate-trigger voltage and gate-trigger current | 4221 | $V_2$ = 6.0 Vdc, $R_L$ = 10 ohms, $R_e$ = 25 ohms maximum | | $V_{GT1}$<br>$I_{GT1}$ | 0.25<br>5 | 3.0<br>70 | Vdc<br>mAdc |
| Forward "on" voltage | 4226 | $I_{TM}$ = 220 A(pk) (pulse), Pulse width = 8.3 ms maximum, Duty cycle = 2% maximum | | $V_{TM}$ | --- | 2.1 | V(pk) |
| Holding current | 4201 | $V_{AA}$ = 22.5 Vdc, $I_{F1}$ = 5 Adc, $I_{F2}$ = 500 mAdc, bias condition D, Gate-trigger source voltage = 6 Vdc, $R_2$ = 50 ohms | | $I_H$ | --- | 40 | mAdc |
| Reverse gate current | 4219 | $V_G$ = -5.0 Vdc | | $I_G$ | --- | -300 | mAdc |
| Subgroup 3 | | | 5 | | | | |
| High-temperature reverse blocking current | 4211 | AC method, $T_C$ = 125°C, bias condition D, f = 60 Hz, $V_{RM}$ = $V_{RRM}$ (see 1.3.1) | | $I_{RRM2}$ | --- | 15 | mA(pk) |
| High-temperature forward blocking current | 4206 | AC method, $T_C$ = 125°C, bias condition D, $V_{DM}$ = $V_{DRM}$ (see 1.3.1) | | $I_{DRM2}$ | --- | 15 | mA(pk) |
| Gate-trigger voltage | 4221 | $T_C$ = 125°C, $R_e$ = 25 ohms maximum, $R_L$ = 500 ohms, $V_2$ = $V_{DRM}$ (see 1.3.1) | | $V_{GT2}$ | 0.25 | --- | Vdc |
| Low-temperature reverse blocking current | 4211 | AC method, $T_C$ = -40°C, bias condition D, f = 60 Hz, $V_{RM}$ = $V_{RRM}$ (see 1.3.1) | | $I_{RRM3}$ | --- | 5 | mA(pk) |
| Low-temperature forward blocking current | 4206 | AC method, $T_C$ = -40°C, bias condition D, $V_{DM}$ = $V_{DRM}$ (see 1.3.1) | | $I_{DRM3}$ | --- | 5 | mA(pk) |
| Low-temperature gate-trigger voltage and gate-trigger current | 4221 | $V_2$ = 6.0 Vdc, $R_L$ = 10 ohms, $R_e$ = 25 ohms maximum, $T_C$ = -40°C | | $V_{GT3}$<br>$I_{GT2}$ | 0.25<br>--- | 3.0<br>130 | Vdc<br>mAdc |



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TABLE I. Group A inspection -Continued.

| Inspection | MIL-STD-750 | | LTPD | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | Method | Conditions | JANTX JANTXV | | Min | Max | |
| Subgroups 4 & 5 | | | | | | | |
| Not applicable | | | | | | | |
| Subgroup 6 | | | 10 | | | | |
| Surge current | 4066 | $I_{TSM}$ = 1000 A(pk) (1/2 sine wave), 10 surges at 1 per minute, $I_O$ = 0, $V_{RM}$ = $V_{RRM}$ (see 1.3.1), $T_C$ = 125°C, surge duration = 7 ms minimum | | | | | |
| Electrical measurements | | See table IV, steps 1, 2, 3, and 4 | | | | | |
| Subgroup 7 | | | 10 | | | | |
| Expotential rate of voltage rise | 4231 | $T_C$ = 125°C, bias condition D, C = .05 µF, $R_L$ = 100 ohms, repetition rate = 60 pps, test duration = 15 s, dv/dt = 200 V/µs | | $V_D$ | | | |
| 2N1910, 2N1792, 2N2031 | | $V_{AA}$ = 50 Vdc | | | 45 | --- | Vdc |
| 2N1911, 2N1793 | | $V_{AA}$ = 100 Vdc | | | 95 | --- | Vdc |
| 2N1913, 2N1795 | | $V_{AA}$ = 200 Vdc | | | 190 | --- | Vdc |
| 2N1915, 2N1797 | | $V_{AA}$ = 300 Vdc | | | 280 | --- | Vdc |
| 2N1916, 2N1798 | | $V_{AA}$ = 400 Vdc | | | 370 | --- | Vdc |
| 2N1805, 2N1799 | | $V_{AA}$ = 500 Vdc | | | 470 | --- | Vdc |
| 2N1805, 2N1800 | | $V_{AA}$ = 600 Vdc | | | 570 | --- | Vdc |
| Circuit-commutated turn-off time | 4224 | $T_C$ = 100°C, $I_{TM}$ = 50 A(pk), $t_{on}$ = 100 ±50 µs, di/dt = 25 A/µs maximum, $V_{RM}$ = $V_{RRM}$ maximum (see 1.3.1) rev. voltage @ $t_1$ = 15 V minimum, repetition rate = 60 pps maximum, dv/dt = 20 V/µs, $V_{DM}$ = $V_{DRM}$ (see 1.3.1), gate bias condition, gate source voltage = 0, gate source resistance = 100 ohms maximum | | $t_q$ | --- | 40 | µs |
| Gate-controlled turn-on time (2N2031 only) | 4223 | $V_{AA}$ = 6 Vdc, $I_{TM}$ = 2A(pk), $V_{GG}$ = 6 Vdc, $t_{p1}$ = 11 µs maximum, $R_e$ = 25 ohms maximum, Forward current pulse = 4A/µs $\leq \frac{di}{dt} \leq$ 200A/µs | | $t_{on}$ | --- | 15 | µs |



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TABLE II. Group B inspection.

| Inspection | MIL-STD-750 | | LTPD |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 15 |
| Resistance to solvents | 1022 | | |
| Subgroup 2 | | | 10 |
| Thermal shock (temperature cycling) | 1051 | Test condition $F_1$, except $T_{Low}$ = -40°C | |
| Hermetic seal<br>Fine leak<br>Gross leak | 1071 | | |
| Electrical measurements | | See table IV, steps 1, 2, 3, and 4 | |
| Subgroup 3 | | | 5 |
| Blocking life | --- | Test conditions per MIL-STD-750, method 1040, test condition A (ac blocking voltage); $T_C$ = 125°C; $V_{RM} = V_{RRM}$ (see 1.3.1), $V_{DM} = V_{DRM}$ (see 1.3.1); $R_{GK}$ = ∞ | |
| Electrical measurement | | See table IV, steps 1, 2, 3, and 4 | |
| Subgroup 4 | | | |
| Decap internal visual design verification | 2075 | | 1 device /0 failure for each lot |
| Subgroup 5 | | | 20 |
| Thermal resistance | 4081 | $I_{T1}$ = 50 Adc, $T_2$ = 125°C maximum, $I_{T2}$ = lowest level which assures complete turn on (approximately 1 Adc), $R_{\theta JC}$ = 0.4 °C/W maximum | |
| Subgroup 6 | | | 10 |
| High-temperature life (non operating) | 1032 | $T_A$ = 150°C, t = 340 hours | |
| Electrical measurements | | See table IV, steps 1, 2, 3, and 4 | |



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TABLE III. Group C inspection.

| Inspection | MIL-STD-750 | | LTPD |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 15 |
| Physical dimensions | 2066 | See figures 1 and 2 | |
| Subgroup 2 | | | 10 |
| Thermal shock (glass strain) | 1056 | Test condition B | |
| Terminal strength stud torque | 2036 | Test condition D2 torque = 150 lb-in, t = 15 seconds minimum | |
| Hermetic seal<br>Fine leak<br>Gross leak | 1071 | | |
| Moisture resistance | 1021 | Omit initial conditioning | |
| External visual | 2071 | | |
| Electrical measurements | | See table IV, steps 1, 2, 3, and 4 | |
| Subgroup 3 | | | 10 |
| Shock | 2016 | 5 blows each $Y_1$, $Y_2$, $Z_1$, axes; 500 G, 1.0 ms. | |
| Vibration, variable frequency | 2056 | 10 G, 100 to 1000 Hz, leads secured or removed | |
| Constant acceleration | 2006 | 2500 G, orientations $X_1$, $Y_1$, $Z_1$ | |
| Electrical measurements | | See table IV, steps 1, 2, 3, and 4 | |
| Subgroup 4 | | | 15 |
| Salt atmosphere (corrosion) | 1041 | | |
| Subgroup 5 | | | 15 |
| Barometric pressure (reduced) | 1001 | 15 mm Hg, $V_{RM} = V_{RRM}$ (see 1.3.1), $V_{DM} = V_{DRM}$ (see 1.3.1), t = 60 s | |
| Subgroup 6 | | | $\lambda = 10$ |
| Intermittent operation life | 1036 | $t_{on}$ = 50 minutes, $t_{off}$ = 10 minutes, see paragraph 4.5.1 | |
| Electrical measurements | | See table IV, steps 1, 2, 3, and 4 | |



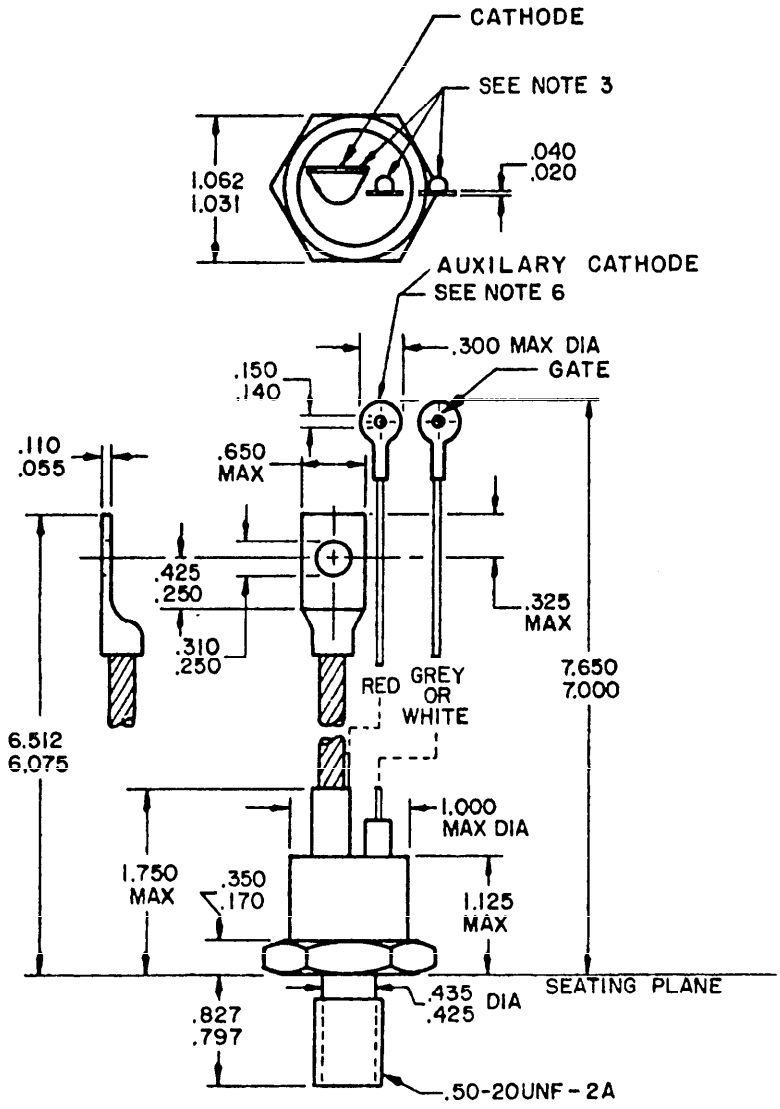
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TABLE IV. Group A, B, and C.

| Step | Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1. | Reverse blocking current | 4211 | AC method, bias condition D, f = 60 Hz, $V_{RM} = V_{RRM}$ (see 1.3.1) | $I_{RRM1}$ | --- | 5 | mA(pk) |
| 2. | Forward blocking current | 4206 | AC method, bias condition D, f = 60 Hz, $V_{DM} = V_{DRM}$ (see 1.3.1) | $I_{DRM1}$ | --- | 5 | mA(pk) |
| 3. | Gate-trigger voltage and gate-trigger current | 4221 | $V_2$ = 6.0 Vdc, $R_L$ = 10 ohms, $R_e$ = 25 ohms maximum | $V_{GT1}$<br>$I_{GT1}$ | 0.25<br>5 | 3.0<br>70 | Vdc<br>mAdc |
| 4. | Forward "on" voltage | 4226 | $I_{TM}$ = 220 A(pk) (pulse), Pulse width = 8.3 ms maximum, Duty cycle = 2% maximum | $V_{TM}$ | --- | 2.1 | V(pk) |



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| INCHES | MM |
|---|---|
| .020 | .51 |
| .040 | 1.02 |
| .055 | 1.40 |
| .110 | 2.79 |
| .140 | 3.56 |
| .150 | 3.81 |
| .170 | 4.32 |
| .250 | 6.35 |
| .300 | 7.62 |
| .310 | 7.87 |
| .325 | 8.26 |
| .350 | 8.89 |
| .425 | 10.80 |
| .435 | 11.05 |
| .50 | 12.70 |
| .650 | 16.51 |
| .797 | 20.24 |
| .827 | 21.01 |
| 1.000 | 25.40 |
| 1.031 | 26.19 |
| 1.062 | 26.97 |
| 1.125 | 28.58 |
| 1.750 | 44.45 |
| 6.075 | 154.31 |
| 6.512 | 165.40 |
| 7.000 | 177.80 |
| 7.650 | 194.31 |

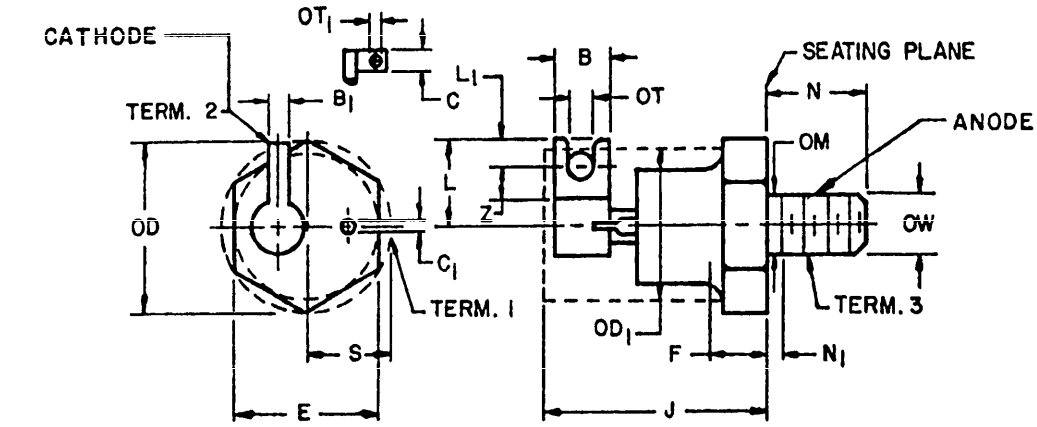
NOTES:

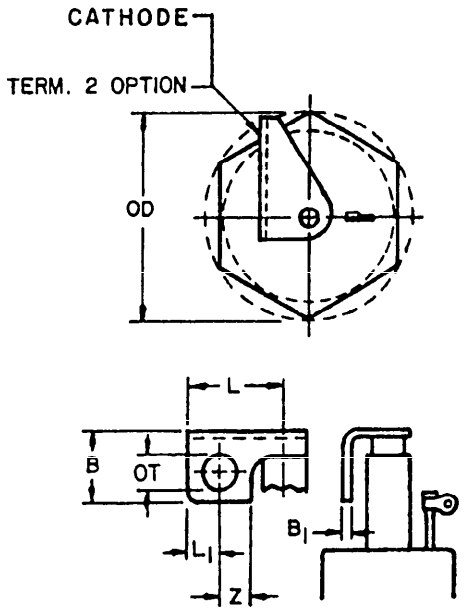
1. Complete threads to extend within 2.50 (63.50 mm) threads of seating plane.
2. Maximum pitch diameter of plated threads shall be basic pitch diameter .4675 (11.87 mm) Ref. (Screw Thread Standards for Federal Services) FED-STD-H28, P1.
3. Angular orientation of these terminals is undefined. Square or radius end of terminal is optional.
4. A chamber or undercut on one or both ends of hex portions is optional.
5. Dimensions are in inches.
6. This auxiliary cathode lead may be cut off if not required for shielding of stray signals which might trigger the thyristor.
7. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Dimensions of controlled rectifier, types 2N1805, 2N1806, 2N1910, 2N1911, 2N1913, 2N1915, 2N1916, and 2N2031, TO-94 outline.



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| SYMBOL | INCHES | | MILLIMETERS | | NOTES |
|---|---|---|---|---|---|
| | MIN | MAX | MIN | MAX | |
| B | .360 | .480 | 9.14 | 12.19 | 1 |
| B1 | .059 | .115 | 1.49 | 2.92 | 1 |
| C | .115 | .160 | 2.92 | 4.06 | 1 |
| C1 | .012 | .050 | .30 | 1.27 | 1 |
| OD | --- | 1.227 | --- | 31.16 | |
| OD1 | --- | 1.031 | --- | 26.19 | 2 |
| E | 1.031 | 1.062 | 26.19 | 26.97 | |
| F | .170 | .400 | 4.32 | 10.16 | 3 |
| J | --- | 1.610 | --- | 45.97 | 2 |
| L | --- | .650 | --- | 16.51 | 2 |
| L1 | .180 | --- | 4.57 | --- | |
| OM | .425 | .499 | 10.80 | 12.67 | 4, 7 |
| N | .797 | .827 | 20.24 | 21.00 | |
| N1 | --- | .125 | --- | 3.18 | 4 |
| S | --- | .575 | --- | 14.60 | 2 |
| OT | .180 | .260 | 4.57 | 6.60 | |
| OT1 | .060 | .080 | 1.52 | 2.03 | |
| OW | .4619 | .4675 | 11.73 | 11.87 | 5 |
| Z | .180 | --- | 4.57 | --- | 6 |

NOTES:

1. Contour and orientation of fixed terminal lugs are undefined.
2. The body and terminals of the device, with the exception of the extended lug length S and L, lies within the cylinder defined by OD1 and length J.
3. A chamfer (or undercut) on one or both ends of the hexagonal portions is optional.
4. Length of incomplete or undercut threads.
5. Pitch diameter of .50-20 UNF 2-A (coated) threads (USA B1. 1-1960).
6. Minimum flat.
7. Complete threads to extend to within 2.50 threads of head.
8. Angular orientation of terminals 1 and 2 is undefined.

FIGURE 2. Dimensions of thyristor types 2N1792, 2N1793, 2N1795, 2N1797, 2N1798, 2N1799, 2N1800, TO-83 outline.



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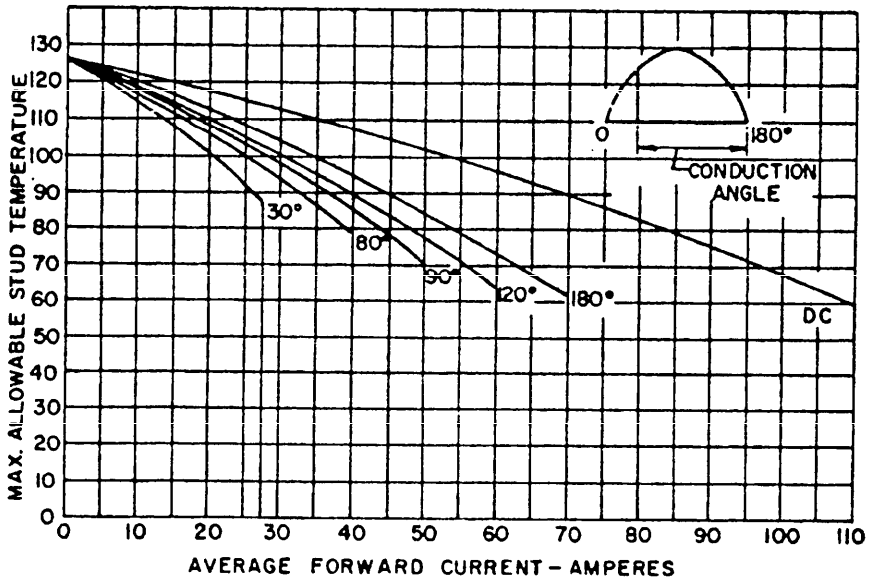



FIGURE 3. Maximum allowable stud temperature.

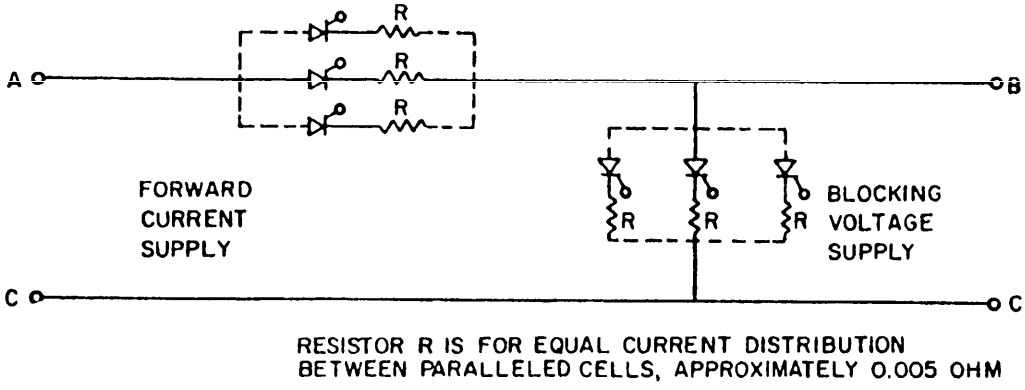



FIGURE 4. Basic test circuit for intermittent life test.



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